

Functions

①

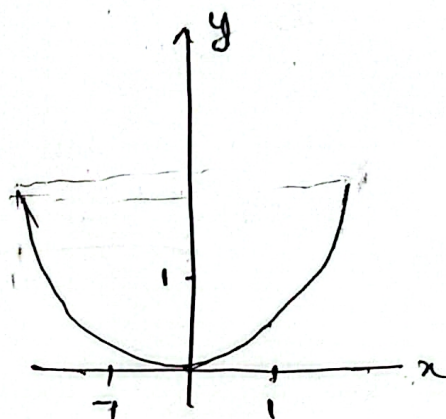
Natural domain consists of all points at which the formula has no divisions by zero and produces only real numbers.

Example

$$w = \sqrt{y - x^2}$$

Domain $y \geq x^2$

Range $[0, \infty)$



The domain of $f(x, y) = \sqrt{y - x^2}$ consists of the shaded region and its bounding parabola $y = x^2$.

Domain $y \geq x^2$

Ex: $w = \frac{1}{xy}$

Domain

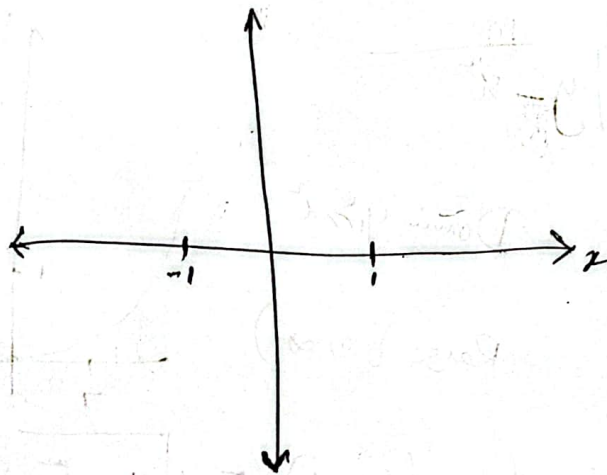
$$xy \neq 0$$

Range $(-\infty, 0) \cup (0, \infty)$

limit of multivariable F_n

Ex: $f(x,y) = \sin^{-1}(x+y)$

Domain of f is the region in which
 $-1 \leq x+y \leq 1$

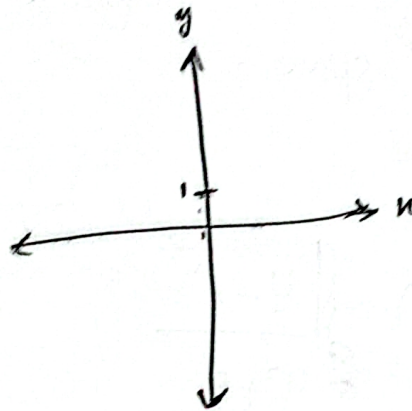


Function	Domain	Range
$w = \sqrt{x^2 + y^2 + z^2}$	Entire 3D space $(-\infty, \infty)$	$[0, \infty)$
$w = \frac{1}{x^2 + y^2 + z^2}$	$(x,y,z) \neq (0,0,0)$	$(0, \infty)$
$w = xy \ln z$	Half space $z > 0$	$(-\infty, \infty)$

Ex: $f(x,y) = xy \sqrt{y-1}$

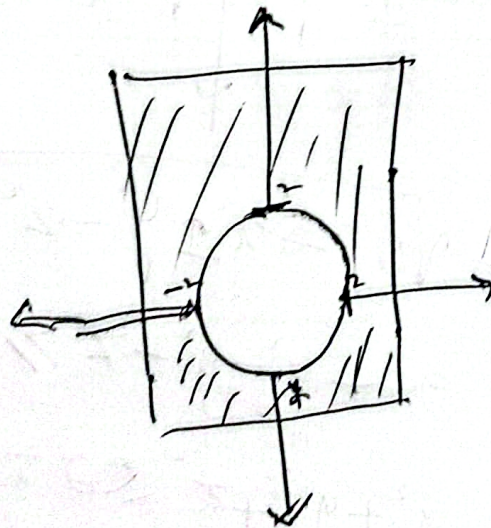
(3)

Domain of f consists of region in xy plane where $y \geq 1$



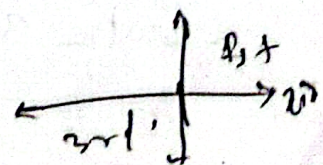
Ex: $f(x,y) = \sqrt{x^2 + y^2 - 4}$

Domain of f consists of region in xy plane where $x^2 + y^2 \geq 4$



Ex: $f(x,y) = \ln xy$
 $xy > 0$

Domain of f consists of region lying in first and third quadrants in xy plane.



Ex: $f(x, y, z) = e^{xyz}$

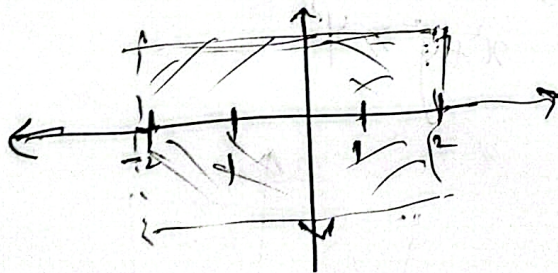
Domain of f consists of region of three of three dimensional space.

Ex: $f(x, y) = \sqrt{\frac{4-x^2}{y^2+3}}$

Domain of f consist of region in xy plane

$$x^2 \leq 4$$

$$-2 \leq x \leq 2$$



Ex: $f(x, y, z) = \sqrt{25 - x^2 - y^2 - z^2}$

$$25 - x^2 - y^2 - z^2 \geq 0$$

$$\text{or } x^2 + y^2 + z^2 \leq 25$$

Domain of f consists of region in three dimensional space occupied by sphere centre at $(0, 0, 0)$ and radius 5.

Concept of limit

⑤

$$f(x) = \lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} \Rightarrow \lim_{x \rightarrow 3} \frac{(x-3)(x+3)}{(x-3)} = \lim_{x \rightarrow 3} x+3 = 3+3 = \underline{6}$$

$$f(x,y) = \frac{x^3 + 2x^2y - xy - 2y^2}{x+2y}$$

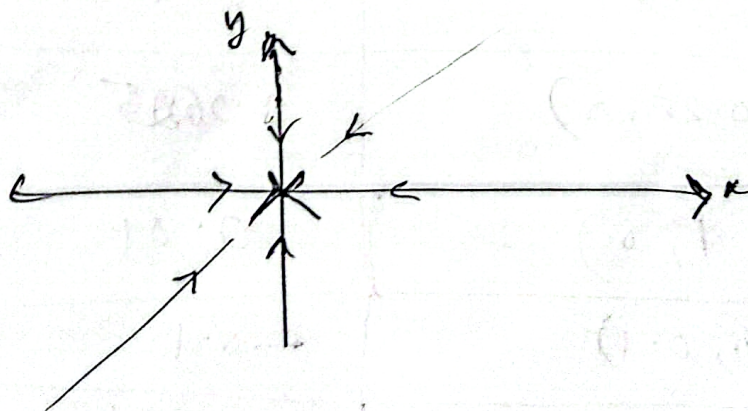
at $x=0, y=0$

$$\Rightarrow \frac{0}{0} \text{ [Indeterminate form]}$$

Functions may be approached from any point in 3D space.

Approaching to $(0,0)$ through x -axis	$f(x,y)$
$(0.5, 0)$	0.25
$(0.25, 0)$	0.0625
$(0.1, 0)$	0.01
$(-0.25, 0)$	0.0625
$(-0.1, 0)$	0.01
$(0, 0.1)$	-0.1
$(0, 0.001)$	-0.001

$(0, 0.00001)$	0.00001
$(0, -0.001)$	0.001
$(0, -0.00001)$	0.00001
Approaching to $(0,0)$ through $y = x$	$f(x,y)$
$(0.5, 0.5)$	-0.25
$(0.1, 0.1)$	-0.09
$(0.01, 0.01)$	-0.0099
$(-0.5, -0.5)$	0.75
$(-0.1, -0.1)$	0.11
$(-0.01, -0.01)$	0.0101



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Ex: $f(x,y) = \frac{xy}{x^2+y^2}$

⑦

$f(0,0)$ is not defined

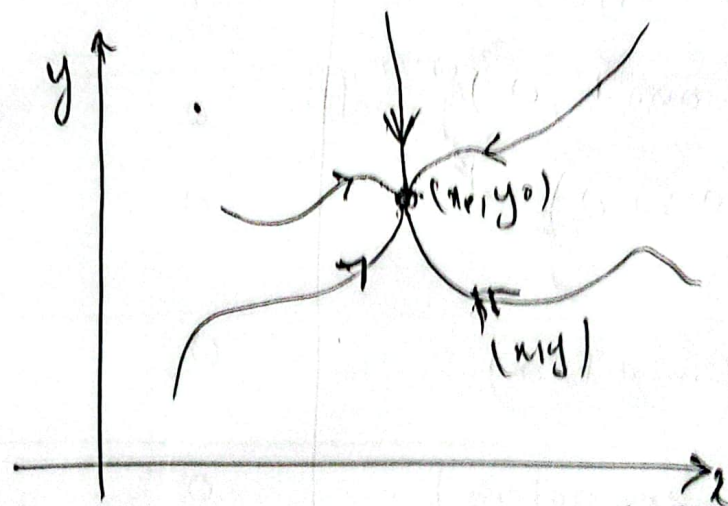
Approaching to $(0,0)$ through x -axis ($y=0$)	$f(x,y)$
$(0.5, 0)$	0
$(0.1, 0)$	0
$(0.01, 0)$	0
$(0.001, 0)$	0
$(0.0001, 0)$	0
$(-0.5, 0)$	0
$(-0.1, 0)$	0
$(-0.001, 0)$	0
$(-0.0001, 0)$	0
Approaching to $(0,0)$ through $y=x$	$f(x,y)$
$(0.5, 0.5)$	0.5
$(0.25, 0.25)$	0.5
$(-0.001, -0.001)$	0.5

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2+y^2} = 0 \text{ (along } y=0 \text{)}$$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2+y^2} = 0.5 \text{ (along } y=x \text{)}$$

therefore Approaching from different ^{paths} we got different function value, therefore limit does not exist.

hence



Example: "through Algebraic method."

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2+y^2}$$

let (x,y) approaching $(0,0)$ along the line $y=x$

$$f(x,y) = \frac{xy}{x^2+y^2} = \frac{x \cdot x}{x^2+x^2} = \frac{1}{1+1} = \frac{1}{2} \Rightarrow x \neq 0$$

$(x,y) \rightarrow (0,0)$ along the line $y=0$

$$f(x,y) = \frac{x \cdot (0)}{x^2 + (0)^2} = 0, x \neq 0$$

②

Thus $f(x,y)$ assume two different values as (x,y) approaches $(0,0)$ along two different paths.

$\lim_{(x,y) \rightarrow (0,0)} f(x,y)$ does not exist.

Rule for Non-existence of a limit

If in $\lim_{(x,y) \rightarrow (a,b)} f(x,y)$

we get two or more different values as we approach (a,b) along different paths then

$\lim_{(x,y) \rightarrow (a,b)} f(x,y)$ does not exist.

The path along which (a,b) is approached may be straight lines or plane curves through (a,b) .

Ex:

lim
(x,y) → (0,0) $\frac{xy}{\sqrt{x^2+y^2}}$

We set: $x = r \cos \theta$, $y = r \sin \theta$

$$\frac{xy}{\sqrt{x^2+y^2}} = \frac{r^2 \cos \theta \sin \theta}{\sqrt{r^2(\cos^2 \theta + \sin^2 \theta)}}$$

$$= \frac{r^2 \cos \theta \sin \theta}{\sqrt{r^2}}$$

$$= \frac{r^2 \cos \theta \sin \theta}{r}$$

$$= r \cos \theta \sin \theta$$

since $r = \sqrt{x^2+y^2}$

As $(x,y) \rightarrow (0,0)$

$$r \rightarrow 0$$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2+y^2}} = \lim_{r \rightarrow 0} r \cos \theta \sin \theta = 0$$

Since $|\cos \theta \sin \theta| \leq 1$ for all values of θ .

$$\lim_{(x,y) \rightarrow (2,1)} \frac{x^3 + 8x^2y - xy - 2y^2}{x + 2y}$$

(1)

$$\lim_{(x,y) \rightarrow (2,1)} (x^3 + 8x^2y - xy - 2y^2)$$

$$\lim_{(x,y) \rightarrow (2,1)} (x + 2y)$$

$$= \frac{8 + 8 - 2 - 2}{2 + 2} = 3$$

Rules for limit

If $\lim_{(x,y) \rightarrow (x_0, y_0)} [cf(x,y)] = cL_1$
 (a) $(x,y) \rightarrow (x_0, y_0)$ (if c is constant)

(b) $\lim_{(x,y) \rightarrow (x_0, y_0)} [f(x,y) + g(x,y)] = L_1 + L_2$

(c) $\lim_{(x,y) \rightarrow (x_0, y_0)} [f(x,y) - g(x,y)] = L_1 - L_2$

(d) $\lim_{(x,y) \rightarrow (x_0, y_0)} [f(x,y) \cdot g(x,y)] = L_1 \cdot L_2$

(e) $\lim_{(x,y) \rightarrow (x_0, y_0)} \left[\frac{f(x,y)}{g(x,y)} \right] = \frac{L_1}{L_2}, (L_2 \neq 0)$

$$\textcircled{f} \lim_{(x,y) \rightarrow (x_0, y_0)} C = C \quad (C \text{ is constant}).$$

$$\textcircled{g} \lim_{(x,y) \rightarrow (x_0, y_0)} x = x_0 \quad \lim_{(x,y) \rightarrow (x_0, y_0)} y = y_0.$$

similarly for the function of three variables.

$\textcircled{h}$ Power Rule: If r and s are integers with no common factors and $s \neq 0$, then

$$\lim_{(x,y) \rightarrow (x_0, y_0)} (f(x,y))^{r/s} = L^{r/s}$$

provided $L^{r/s}$ is real number. (If s is even, we assume that $L > 0$.)

Find:

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$$\lim_{(x,y) \rightarrow (0,1)} \frac{x - xy + 3}{x^2y + 5xy - y^3}$$

$$\lim_{(x,y) \rightarrow (0,0)} \left[x - xy + 3 \right]$$

$$\lim_{(x,y) \rightarrow (0,0)} \left[x^2y + 5xy - y^3 \right]$$

$$\lim_{(x,y) \rightarrow (0,0)} \left[x \right] - \lim_{(x,y) \rightarrow (0,0)} \left[xy \right] + \lim_{(x,y) \rightarrow (0,0)} \left[3 \right]$$

$$\lim_{(x,y) \rightarrow (0,0)} \left[x \right] - \lim_{(x,y) \rightarrow (0,0)} \left[5xy \right] + \lim_{(x,y) \rightarrow (0,0)} \left[y^3 \right]$$

$$= \frac{0 - (0(1) + 3)}{0 - 5(0)(1) - (-1)^3} = \frac{3}{-1} = -3.$$

Example

$\lim_{(x,y) \rightarrow (0,0)}$

$$\frac{x^2 - xy}{\sqrt{x} - \sqrt{y}}$$

form
dir

coefficient

$\lim_{(x,y) \rightarrow (0,0)}$

$$\frac{x(x-y)}{\sqrt{x} - \sqrt{y}}$$

Now

$\lim_{(x,y) \rightarrow (0,0)}$

$$\frac{x(x-y)}{\sqrt{x} - \sqrt{y}}$$

$$\times \frac{\sqrt{x} + \sqrt{y}}{\sqrt{x} + \sqrt{y}}$$

$\lim_{(x,y) \rightarrow (0,0)}$

$$\frac{x(x-y)(\sqrt{x} + \sqrt{y})}{(x-y)}$$

$\lim_{(x,y) \rightarrow (0,0)}$

$$x(\sqrt{x} + \sqrt{y}) = 0(\sqrt{0} + \sqrt{0})$$
$$= 0(0+0)$$
$$= 0$$

Example: Find $\lim_{(x,y) \rightarrow (0,0)} \frac{4xy^2}{x^2+y^2}$ if it exists. (15)

at x -axis; $y=0, x \neq 0$

$$\frac{4xy^2}{x^2+y^2} = \frac{0}{x^2+0} = 0$$

at y -axis; $x=0, y \neq 0$

$$\frac{4xy^2}{x^2+y^2} = \frac{0}{(0)^2+y^2} = 0$$

Example: Find the limit

$$\lim_{\substack{(x,y) \rightarrow (1,1) \\ x \neq y}} \frac{x^2 - 2xy + y^2}{x-y}$$

$$\lim_{\substack{(x,y) \rightarrow (1,1) \\ x \neq y}} \frac{(x-y)^2}{(x-y)} = \lim_{\substack{(x,y) \rightarrow (1,1) \\ x \neq y}} (x-y) = 0$$

Example: Find the limit

$$\lim_{\substack{(x,y) \rightarrow (1,1) \\ x \neq 1}} \frac{xy - y - 2x + 2}{x-1}$$

$$\lim_{\substack{(x,y) \rightarrow (1,1) \\ x \neq 1}} \frac{y(x-1) - 2(x-1)}{(x-1)} = \lim_{\substack{(x,y) \rightarrow (1,1) \\ x \neq 1}} \frac{(x-1)(y-2)}{(x-1)}$$

$$\lim_{(x,y) \rightarrow (1,1)} (y-2) = 1-2 = -1$$

Formula
test

Example: Find the limit

$$\lim_{(x,y) \rightarrow (2,-4)} \frac{y+4}{x^2y - xy + 4x^2 - 4x}$$

$y \neq -4, x \neq x^2$

$$\lim_{(x,y) \rightarrow (2,-4)} \frac{y+4}{x^2y + 4x^2 - (xy + 4x)}$$

$y \neq -4, x \neq x^2$

$$\lim_{(x,y) \rightarrow (2,-4)} \frac{y+4}{x^2(y+4) - x(y+4)}$$

$y \neq -4, x \neq x^2$

$$\lim_{(x,y) \rightarrow (2,-4)} \frac{(y+4)}{(y+4)(x^2-x)} = \frac{1}{x^2-x}$$

$y \neq -4, x \neq x^2$

$$f(x,y) = \lim_{(x,y) \rightarrow (2,-4)} \frac{1}{x^2-x}$$

$y \neq -4, x \neq x^2$

$$= \frac{1}{2}$$

∴ find the limit

(2)

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ x \neq y}} \frac{x - y + 2\sqrt{x} - 2\sqrt{y}}{\sqrt{x} - \sqrt{y}}$$

$$f(x,y) = \frac{x - y + 2\sqrt{x} - 2\sqrt{y}}{\sqrt{x} - \sqrt{y}}$$

$$= \frac{x - y}{\sqrt{x} - \sqrt{y}} + \frac{2\sqrt{x} - 2\sqrt{y}}{\sqrt{x} - \sqrt{y}}$$

$$= \frac{x - y}{\sqrt{x} - \sqrt{y}} + 2 \frac{(\cancel{\sqrt{x} - \sqrt{y}})}{(\cancel{\sqrt{x} - \sqrt{y}})}$$

$$= \frac{x - y}{\sqrt{x} - \sqrt{y}} + 2$$

$$= \frac{(\cancel{\sqrt{x} - \sqrt{y}})(\sqrt{x} + \sqrt{y})}{(\cancel{\sqrt{x} - \sqrt{y}})} + 2$$

$$\lim_{\substack{(x,y) \rightarrow (0,0) \\ x \neq y}}$$

$$= \sqrt{x} + \sqrt{y} + 2$$

$$= \sqrt{0} + \sqrt{0} + 2 = 2 \underline{A.}$$

Ex: Find the limit

$$\lim_{\substack{(x,y) \rightarrow (4,3) \\ x \neq y+1}} \frac{\sqrt{x} - \sqrt{y+1}}{x - y - 1}$$

$$\begin{aligned} f(x,y) &= \frac{\sqrt{x} - \sqrt{y+1}}{x - (y+1)} = \frac{\sqrt{x} - \sqrt{y+1}}{(\sqrt{x} - \sqrt{y+1})(\sqrt{x} + \sqrt{y+1})} \\ &= \frac{1}{\sqrt{x} + \sqrt{y+1}} \end{aligned}$$

$$\begin{aligned} \lim_{\substack{(x,y) \rightarrow (4,3) \\ x \neq y+1}} &= \frac{1}{\sqrt{4} + \sqrt{3+1}} = \frac{1}{\sqrt{4} + \sqrt{4}} \\ &= \frac{1}{2+2} = \frac{1}{4} \end{aligned}$$